

演習1

$$L(V, \Lambda) = \text{tr}(V^T X^T X V) - \text{tr} \left[\Lambda (V^T V - I_k) \right]$$

$$\frac{\partial L}{\partial V} = 2X^T X V - \frac{\partial}{\partial V} \text{tr} \left[(V \Lambda V^T) \right]$$

2.2.2 公式: $\frac{\partial}{\partial X} \text{tr}(X^T M X) = (M + M^T) X$ について.

$Y = X^T$ とおく

$$\frac{\partial}{\partial Y^T} \text{tr}(Y M Y^T) = (M + M^T) Y^T$$

ゆえに $\frac{\partial}{\partial Y} \text{tr}(Y M Y^T) = \left[(M + M^T) Y^T \right]^T = Y (M^T + M)$

Y を V , M を Λ に置換して

$$\frac{\partial}{\partial V} \text{tr} \left[(V \Lambda V^T) \right] = V (\Lambda + \Lambda^T) = 2V\Lambda$$

よって $\frac{\partial L}{\partial V} = 2X^T X V - 2V\Lambda$

右辺 = 0 として $X^T X V = V\Lambda$

演習2

$$\begin{aligned}
 \|X - XVV^T\|^2 &= \text{tr} \left[(X - XVV^T)^T (X - XVV^T) \right] \\
 &= \text{tr} \left[(X^T - VV^T X^T) (X - XVV^T) \right] \\
 &= \text{tr} \left(X^T X - X^T X V V^T - V V^T X^T X + V V^T X^T X V V^T \right) \\
 &= \text{tr} \left(X^T X - V^T X^T X V \right) \quad (\because V^T V = I_k)
 \end{aligned}$$

$X^T X$ は V によらないから

$$\min \text{tr} (X^T X - V^T X^T X V) \Leftrightarrow \max (V^T X^T X V)$$

よってこの問題は

$$\max_{V \in \mathbb{R}^{n \times k}} (V^T X^T X V) \quad \text{s.t. } V^T V = I_k \quad \text{の PCA 問題と同等}$$

演習3

$$\begin{aligned}
 \Pr(|X - \mu| \geq a) &= \Pr((X - \mu)^2 \geq a^2) \quad (\because a > 0) \\
 &\leq \frac{\mathbb{E}[(X - \mu)^2]}{a^2} \quad \leftarrow \text{Markov の不等式} \\
 &= \frac{\sigma^2}{a^2}
 \end{aligned}$$

演習4

$$\begin{aligned}
 1. \ E[u^T v] &= E\left[\sum_{i=1}^n (u_i v_i)\right] \\
 &= \sum_{i=1}^n \left(E[u_i v_i]\right) \\
 &= \sum_i \left(E[u_i] E[v_i]\right) \\
 &= 0
 \end{aligned}$$

$$(0 \ u \ 0) \begin{pmatrix} 1 \\ v \\ 1 \end{pmatrix} = 0 \cdot 1 + 0 \cdot v + 0 \cdot 1 = 0$$

$$\begin{aligned}
 \text{Var}[u^T v] &= \text{Var}\left[\sum_i (u_i v_i)\right] \\
 &= \sum_i \left(\text{Var}[u_i v_i]\right) \\
 &= \sum_i \left(E[u_i^2 v_i^2] - (E[u_i v_i])^2\right) \\
 &= \sum_i \left(E[u_i^2] E[v_i^2]\right) \\
 &= n \left(\frac{1}{n} \cdot \frac{1}{n}\right) = \frac{1}{n}
 \end{aligned}$$

$$2. \ P_r(|u^T v| \geq \epsilon) = P_r(|u^T v - 0| \geq \epsilon) \leq \frac{\frac{1}{n}}{\epsilon} = \frac{1}{n\epsilon}$$

演習5

$$1. E[\|u\|^2] = E[u^T u] \\ = E\left[\sum_i u_i^2\right] = \sum_i \left(E[u_i^2]\right) = n \cdot \frac{1}{n} = 1$$

$$\text{Var}[\|u\|^2] = \text{Var}\left[\sum_i u_i^2\right] \\ = \sum_i \left(\text{Var}[u_i^2]\right) \\ = \sum_i \left(E[u_i^4] - (E[u_i^2])^2\right) \\ = \sum_i \left(3 \cdot \left(\frac{1}{n}\right)^2 - \left(\frac{1}{n}\right)^2\right) = n \cdot 2 \left(\frac{1}{n}\right)^2 = \frac{2}{n}$$

$$2. |\|u\|^2 - 1| \geq \varepsilon \Leftrightarrow \|u\|^2 \geq 1 + \varepsilon \text{ または } \|u\|^2 \leq 1 - \varepsilon$$

$$\Pr(|\|u\|^2 - 1| \geq \varepsilon) = \Pr(\|u\|^2 \geq 1 + \varepsilon) + \Pr(\|u\|^2 \leq 1 - \varepsilon) \quad (\because \varepsilon > 0)$$

$$u_i \sim N\left(0, \frac{1}{n}\right) \text{ より } Z_i = \sqrt{n} u_i \sim N(0, 1)$$

$$S = \sum_i Z_i^2 \text{ とおす}$$

$$\Pr(S \geq (1 + \varepsilon)n) = \Pr\left(\sum_i (\sqrt{n} u_i)^2 \geq (1 + \varepsilon)n\right)$$

$$= \Pr\left(\sum_i u_i^2 \geq 1 + \varepsilon\right)$$

$$= \Pr(\|u\|^2 \geq 1 + \varepsilon) \leq \exp\left(-\frac{n}{2}(\varepsilon - \ln(1 + \varepsilon))\right)$$

$$\Pr(S \leq (1 - \varepsilon)n) = \Pr(\|u\|^2 \leq 1 - \varepsilon) \leq \exp\left(-\frac{n}{2}(-\varepsilon - \ln(1 - \varepsilon))\right)$$

$$\text{したがって } \Pr(|\|u\|^2 - 1| \geq \varepsilon) \leq \exp\left(-\frac{n}{2}(\varepsilon - \ln(1 + \varepsilon))\right) + \exp\left(-\frac{n}{2}(-\varepsilon - \ln(1 - \varepsilon))\right)$$